Math 141, S06 **Name:**

Exam 3 **Section:** **SSN: xxx-xx-**

<u>**Instructions:**</u>

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphic calculator is allowed in the exam.

Problem	Points	Score
1	18	
2	16	
3	12	
4	15	
5	25	
6	14	
Total	100	

Good Luck !

Problem 1. (18 points) Let $f(x) = x^4 + \frac{4}{3}x^3 - 12x^2$.

(a) Find the intervals on which f is increasing and the intervals on which f is decreasing.

Solution: $f'(x) = 4x^3 + 4x^2 - 24x = 4x(x^2+x-6) = 4x(x+3)(x-2)$

$f'(x) = 0 \Rightarrow 4x(x+3)(x-2) = 0 \Rightarrow x = -3, 0, 2$.

Intervals	$f'(x)$	conclusion
$x < -3$	< 0	decreasing
$-3 < x < 0$	> 0	increasing
$0 < x < 2$	< 0	decreasing
$x > 2$	> 0	increasing

(b) Find the open intervals on which f is concave up and on which f is concave down.

Solution: $f''(x) = 12x^2 + 8x - 24 = 4(3x^2+2x-6)$

$f''(x) = 0 \Rightarrow 4(3x^2+2x-6) = 0 \Rightarrow x = \frac{-1-\sqrt{19}}{3}, \frac{-1+\sqrt{19}}{3}$

intervals	$f''(x)$	conclusion
$x < \frac{-1-\sqrt{19}}{3}$	> 0	Concave up
$\frac{-1-\sqrt{19}}{3} < x < \frac{-1+\sqrt{19}}{3}$	< 0	concave down
$x < \frac{-1+\sqrt{19}}{3}$	> 0	concave up

(c) Find all relative extrema of f using the **second derivative test.**

Solution: From (a), we know f has three critical points $x = -3, 0, 2$.

i) $x = -3$, $f''(-3) = 4(3\cdot 9 - 6 - 6) = 60 > 0$, so

$f(-3) = (-3)^4 + \frac{4}{3}(-3)^3 - 12(-3)^2 = 81 - 36 - 108 = -63$

is a local minimum.

ii) $x = 0$, $f''(0) = 4(0+0-6) = -24 < 0$, so

$f(0) = 0$ is a local maximum.

iii) $x = 2$, $f''(2) = 4(12+4-6) = 40 > 0$, so

$f(2) = 2^4 + \frac{4}{3}\cdot 2^3 - 12\cdot 2^2 = -\frac{64}{3}$ is a local minimum.

Problem 2. (16 points) Define $f(x) = x^{\frac{5}{3}} - 20x^{\frac{2}{3}}$.

(a) Find all critical points of f, and for each critical point, determine whether it is a stationary point or not.

Solution. $f'(x) = \frac{5}{3}x^{\frac{2}{3}} - \frac{40}{3}x^{-\frac{1}{3}} = \frac{5}{3}x^{-\frac{1}{3}}(x-8)$

$= \frac{5(x-8)}{3x^{\frac{1}{3}}}$

i) When $x=8$, $f'(x)=0$, so $x=8$ is a critical point, and is in fact a stationary point

ii) When $x=0$, $f'(x)$ is not defined, so $x=0$ is a critical point, but it is not a stationary point.

(b) Find the absolute extrema of f, if any, on the closed interval $[-1, 20]$.

Solution: Evaluate f at the critical points $0, 8$ and two endpoint $-1, 20$, we get.

$f(0) = 0$

$f(8) = 32 - 20 \cdot 4 = -48$

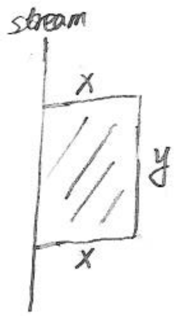
$f(-1) = (-1)^{\frac{5}{3}} - 20 \cdot (-1)^{\frac{2}{3}} = -1 - 20 = -21$

$f(20) = 20^{\frac{5}{3}} - 20 \cdot 20^{\frac{2}{3}} = 0$

So f has an absolute maximum 0 at two points $x = 0, 20$ and an absolute minimum -48 at $x=8$ over the interval $[-1, 20]$.

Problem 3. (12 points) A rectangle field is to be bounded by a fence on three sides and by a straight stream on the fourth side. Find the dimensions of the field with maximum area that can be enclosed with 1000 feet of fence. (Hint: Formulate the problem as a max/min problem on an appropriate interval.)

Solution: The area of the field



$$A = xy$$

where $2x + y = 1000$

So we have $y = 1000 - 2x$, then

$$A = x \cdot (1000 - 2x)$$
$$= -2x^2 + 1000x.$$

This problem can be formulated as a maximization problem of A for $0 \le x \le 500$ (Since $x \geq 0$ and $2x \le 1000$).

$$\frac{dA}{dx} = -4x + 1000 = 0$$
$$\Rightarrow x = 250 \quad \Rightarrow y = 500$$

Thus, the maximum of the area of the field is

$$A = 250 \cdot 500$$
$$= 125000 \text{ feet}^2$$

Problem 4. (15 points) Let $s(t) = \dfrac{t}{t^2+4}$ be the position of a particle moving along a coordinate line, where s is in feet and t is in seconds ($t \geq 0$).

(a) Find the velocity function $v(t)$ for the particle.

$$V(t) = S'(t) = \frac{1\cdot(t^2+4) - t(2t)}{(t^2+4)^2}$$

$$= \frac{4-t^2}{(t^2+4)^2} \text{ feet/second}$$

(b) Find the acceleration function $a(t)$ of this particle.

$$a(t) = V'(t) = \frac{(-2t)(t^2+4)^2 - (4-t^2)\cdot 2(t^2+4)\cdot 2t}{(t^2+4)^4}$$

$$= \frac{-2t\cancel{(t^2+4)}[(t^2+4)+2(4-t^2)]}{(t^2+4)^{\cancel{4}\,3}}$$

$$= \frac{-2t(12-t^2)}{(t^2+4)^3}$$

(c) Find the position of the particle when it attains its maximum speed.

The particle attains its maximum speed at $a(t)=0$.

Thus

$$-2t(12-t^2)=0$$

$$t = 0,\ 2\sqrt{3},\ -2\sqrt{3}$$

Since $t \geqslant 0$, we evaluate $V(t)$ at $t=0, 2\sqrt{3}$

$$|V(0)| = 1$$

$$|V(2\sqrt{3})| = \left|\frac{4-12}{(12+4)^2}\right| = \frac{1}{32}$$

So the particle attains its maximum speed at $t=0$, and its position is

$$S(0) = 0.$$

Problem 5. (25 points) Evaluate each of the following indefinite integrals.

(a) $\int (x^7 - x^{-3} - 2\sqrt[3]{x})\, dx$

$= \frac{x^8}{8} + \frac{x^{-2}}{2} - 2 \cdot \frac{3}{4} x^{\frac{4}{3}} + C$

$= \frac{x^8}{8} + \frac{1}{2x^2} - \frac{3}{2} x^{\frac{4}{3}} + C$

(b) $\int 2x(x^2-2)^9\, dx \underset{du=2xdx}{\overset{u=x^2}{=}} \int (u-2)^9 du = \frac{(u-2)^{10}}{10} + C$

$= \frac{(x^2-2)^{10}}{10} + C$

(c) $\int \frac{1}{y \ln y}\, dy \underset{du=\frac{1}{y}dy}{\overset{u=\ln y}{=}} \int \frac{1}{u} du = \ln|u| + C$

$= \ln|\ln y| + C$

(d) $\int \frac{dx}{e^x} = \int e^{-x} dx = -e^{-x} + C$

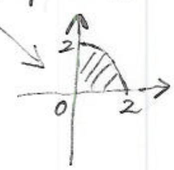
(e) $\int \frac{\cos t}{1+\sin^2 t} dt \underset{du=\cos t dt}{\overset{u=\sin t}{=}} \int \frac{1}{1+u^2} du = \tan^{-1} u + C$

$= \tan^{-1}(\sin t) + C$

Problem 6. (14 points)
(a) Use properties of the definite integral and appropriate geometric formula to evaluate

$$\int_0^2 5 + 3\sqrt{4 - x^2}\, dx.$$

Solution: Notice that $y = \sqrt{4-x^2}$, $0 \leq x \leq 2$ represent quarter of the circle $x^2 + y^2 = 2^2$



$$\int_0^2 5 + 3\sqrt{4-x^2}\, dx$$
$$= \int_0^2 5\, dx + 3\int_0^2 \sqrt{4-x^2}\, dx$$
$$= 5 \cdot 2 + 3 \cdot \frac{\pi \cdot 2^2}{4}$$
$$= 10 + 3\pi$$

(b) Suppose $\int_0^3 f(x)\, dx = -2$, $\int_2^3 f(x)\, dx = -4$, and $\int_0^2 g(x)\, dx = 3$. Find

$$\int_0^2 f(x) + 2g(x)\, dx.$$

Solution: $\int_0^2 f(x)\, dx + \int_2^3 f(x)\, dx = \int_0^3 f(x) dx$

$$\int_0^2 f(x) dx + (-4) = (-2)$$
$$\int_0^2 f(x) = -2 + 4 = 2$$

So $\int_0^2 f(x) + 2g(x) dx = \int_0^2 f(x) dx + 2\int_0^2 g(x) dx$

$$= 2 + 2 \cdot 3$$
$$= 8$$